

AWS RDS (MariaDB) SETUP

Prepared By: Kritika Agarwal

Project: AWS RDS Setup

Objective

- Login AWS
- Project Purpose

Architecture Diagram

- Structure
- Explanation

Tools & Services Used

- AWS RDS
- MariaDB
- Amazon EC2
- Security Group
- Linux Terminal

Implementation Steps

- Command used
- Output

Conclusion

- Summary
- Learning Outcome

1. OBJECTIVE-

- Login to AWS Console.
- Search RDS.
- Open the RDS Dashboard

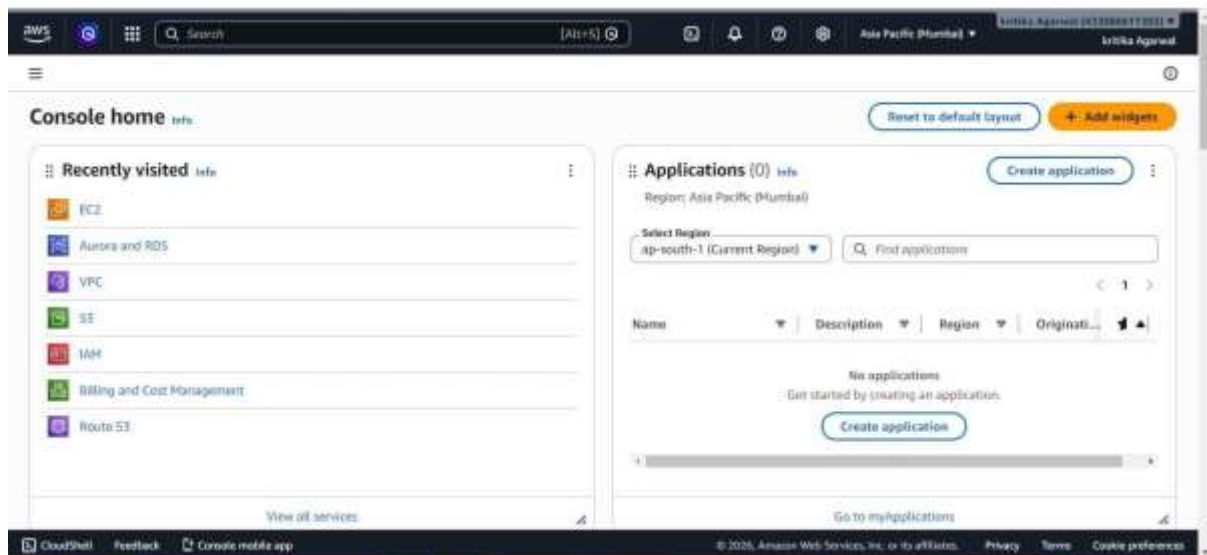


FIG: AWS CONSOLE HOME PAGE

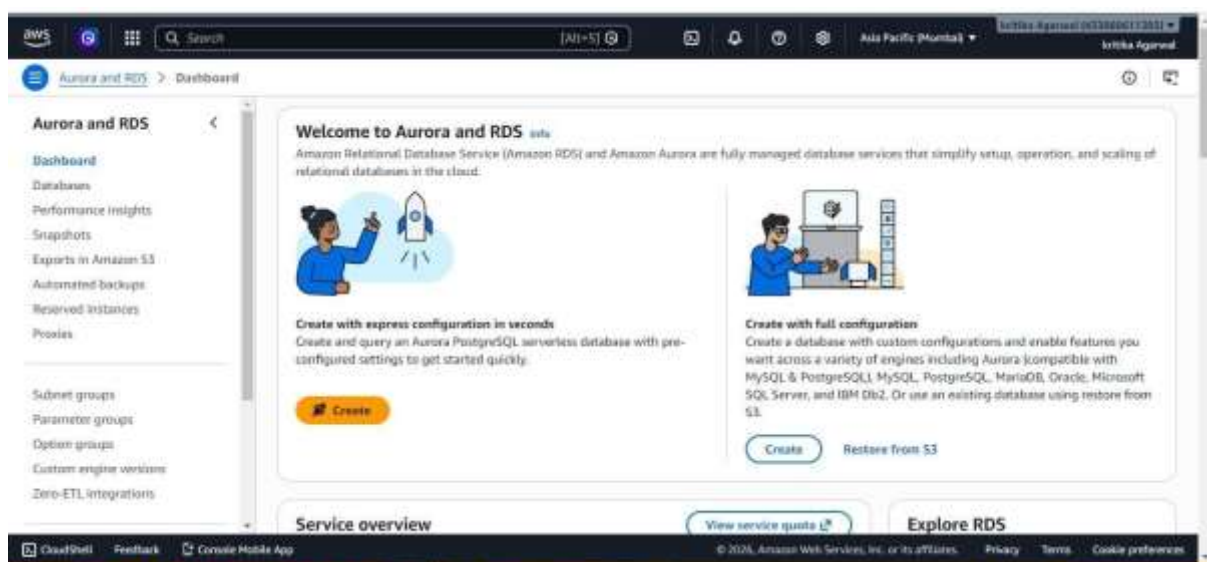


FIG:RDS DASHBOARD

2. ARCHITECTURE DIAGRAM-

1. User:

- User AWS Management Console me login karta hai.
- Yahin se RDS service ko manage karta hai.

2. AWS Management Console:

- AWS ka web interface hai.
- Iske through naya RDS instance create aur configure kiya jata hai.

3. Amazon RDS (MariaDB):

- AWS ki managed database service hai.

- Ye database setup, backup, monitoring aur maintenance automatically handle karti hai.

4. RDS Database Instance:

- Ye actual MariaDB server hota hai.
- Isme database create hota hai aur data store hota hai.

5. Database (Tables):

- Database ke andar tables hoti hain.
- Tables me application ka data rows aur columns ke form me save hota hai.

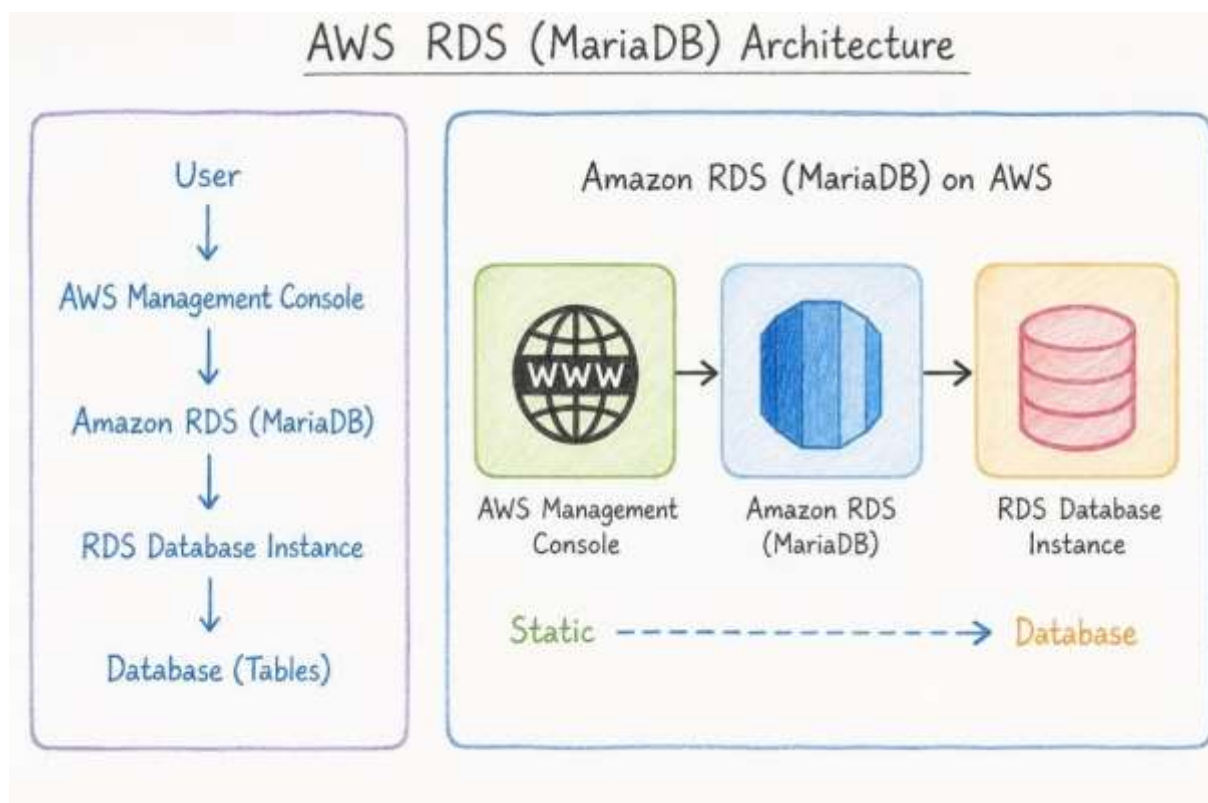


FIG: RDS ARCHITECTURE

3. TOOLS & SERVICES USED-

STEP1: Launch EC2 Instance-

- Open EC2 Dashboard.
- Launch EC2 Instance.
- Select Amazon Linux.
- Create key Pair.
- Security Group.
- EC2 instance running.

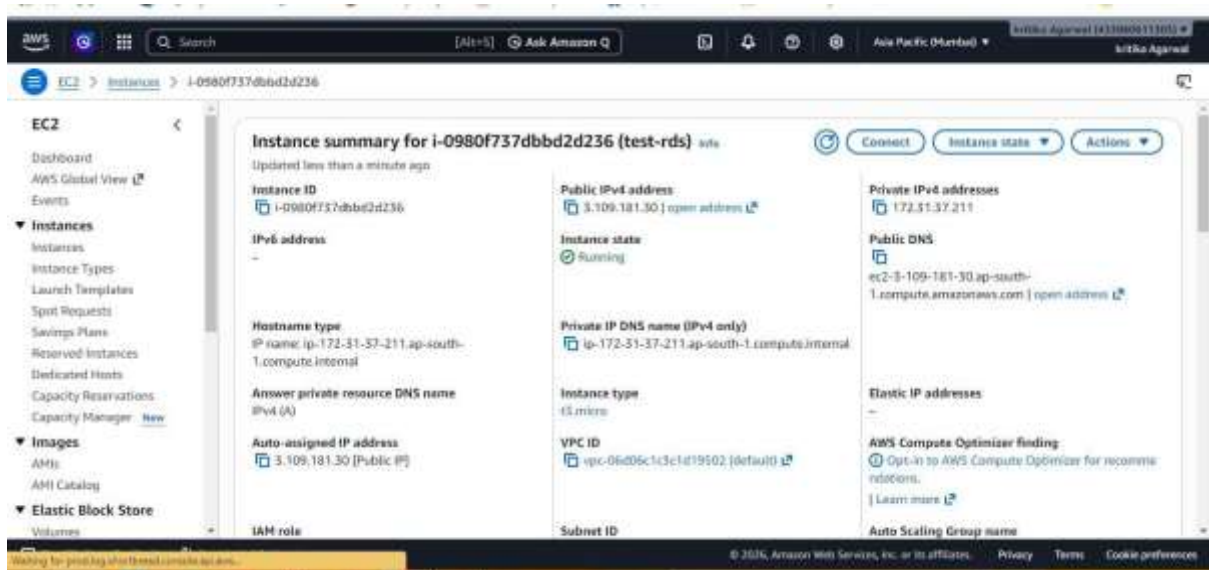


FIG:INSTANCE RUNNING

STEP2:Open RDS-

- Login to AWS Console
- Search RDS
- Click Create Database

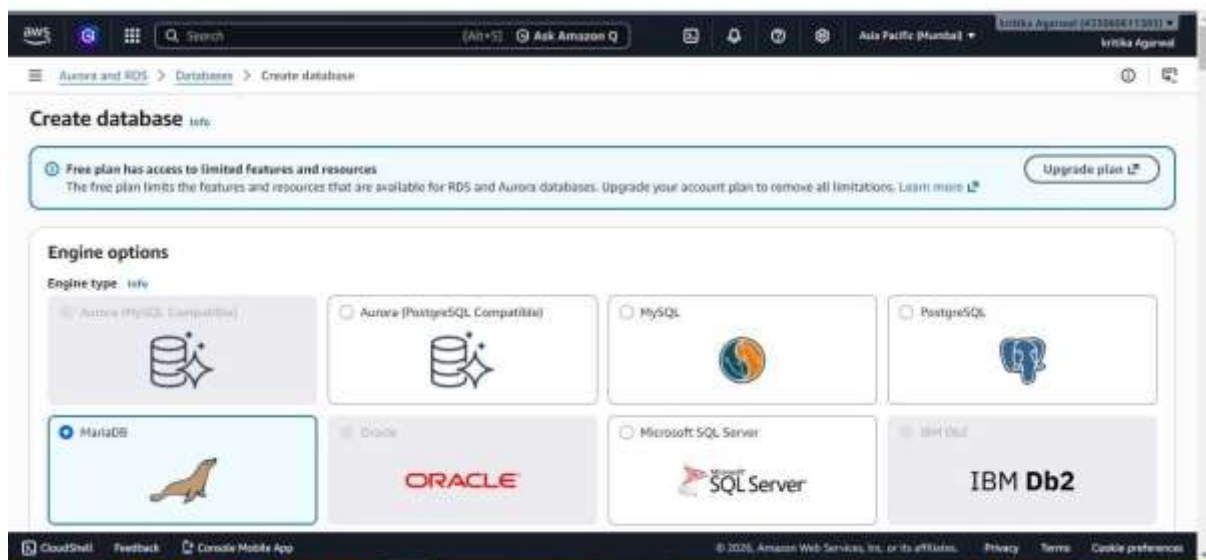


FIG: CREATE DATABASE

STEP3: Choose Database Creation-

- Select Standard Create.
- Engine Type → MariaDB.
- Choose the latest/default version.

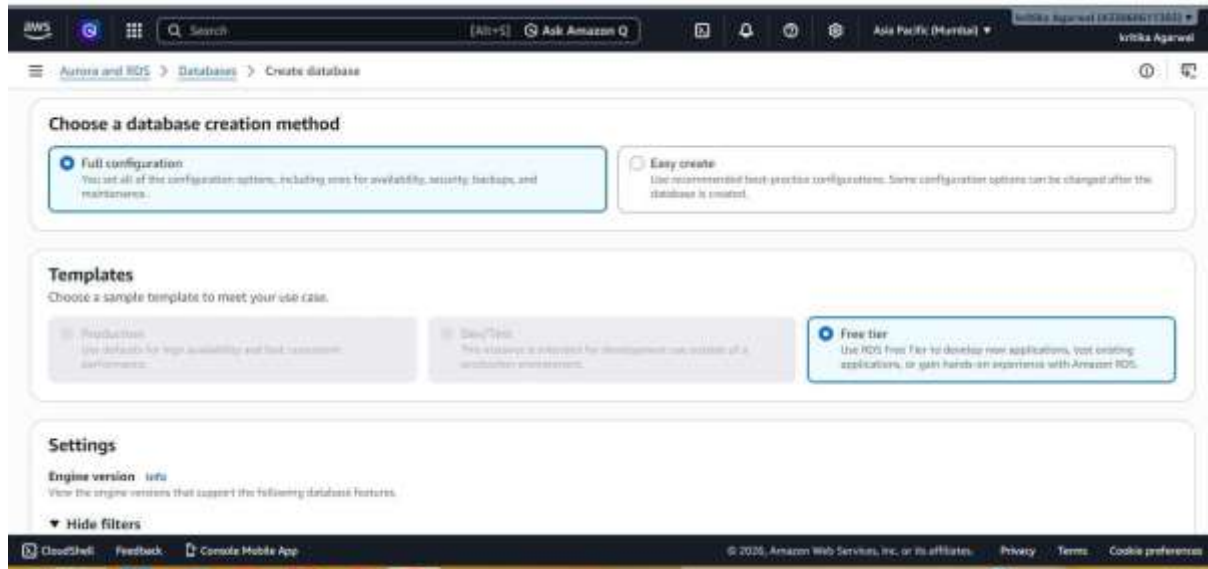


FIG: CHOOSE DATABASE

STEP4: Configure Database-

- DB Instance Identifier → database01
- Master Username → admin
- Create a strong password.

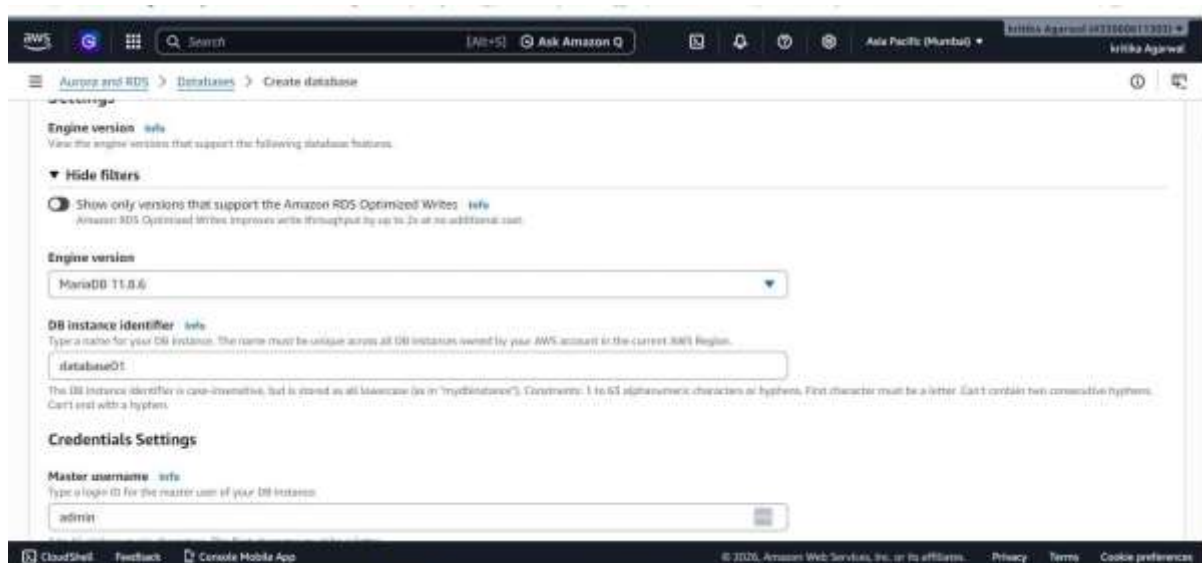


FIG-1:DB IDENTIFIER

The screenshot shows the 'Create database' page in the AWS Management Console. The 'Master username' field is set to 'admin'. Under 'Credentials management', 'Managed in AWS Secrets Manager - most secure' is selected. The 'Master password' field is masked, and a 'Password strength' bar is shown. The 'Confirm master password' field is also masked. The page includes a top navigation bar with the AWS logo, search bar, and user profile, and a bottom footer with links to CloudShell, Feedback, and Console Mobile App.

FIG:MASTER USERNAME

STEP5:Instance Configuration-

- DB Instance Class → db.t3.micro
- Storage → Default (20 GB).

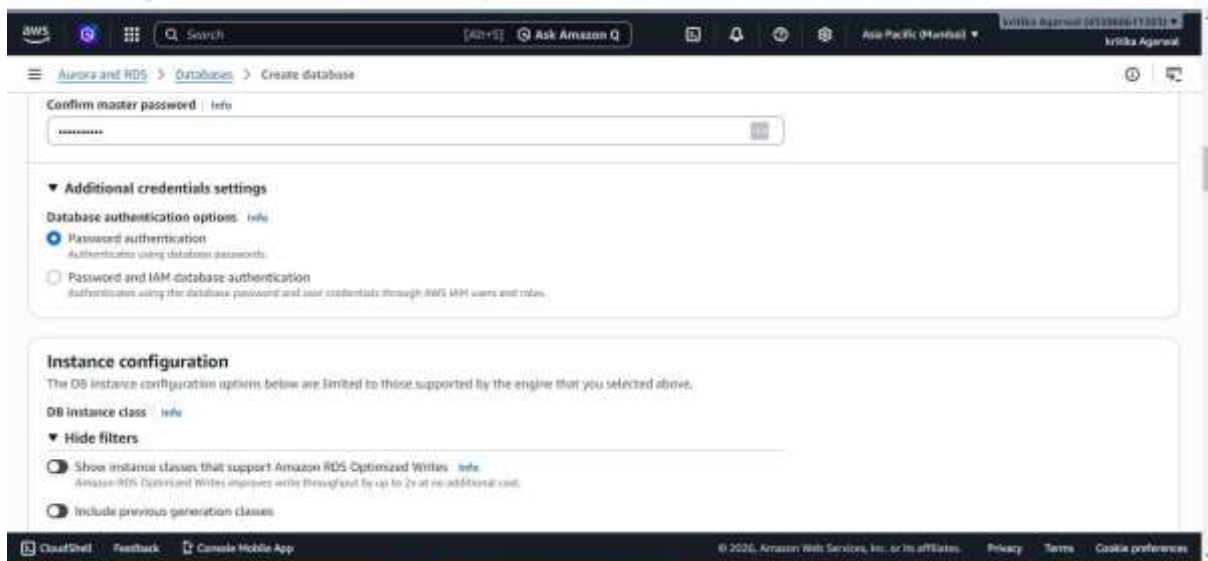


FIG-1:INSTANCE CONFIGURATION

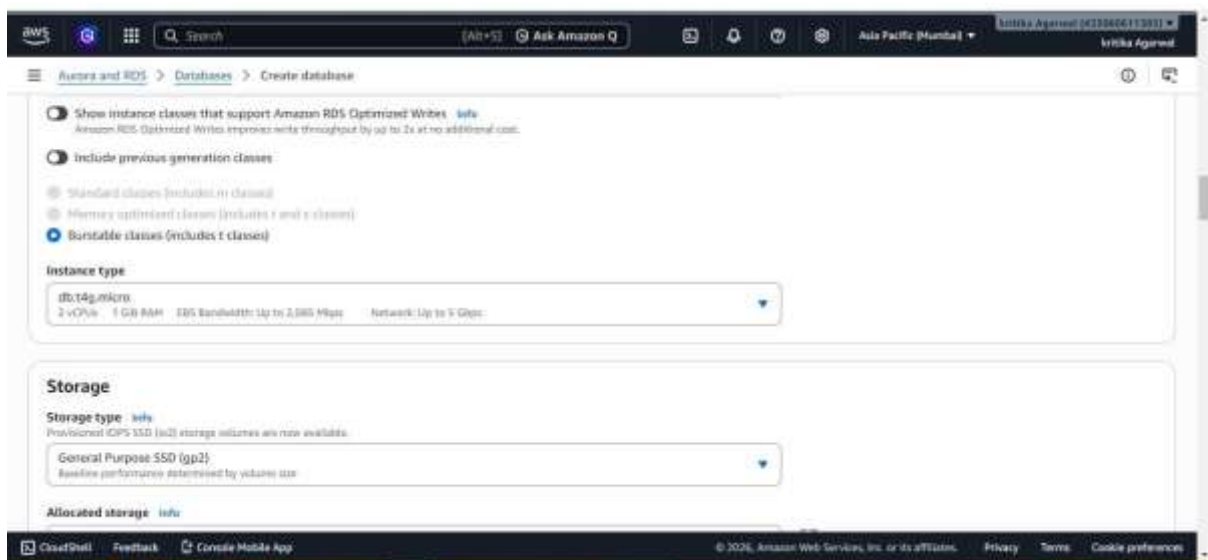


FIG-2: INSTANCE CLASS

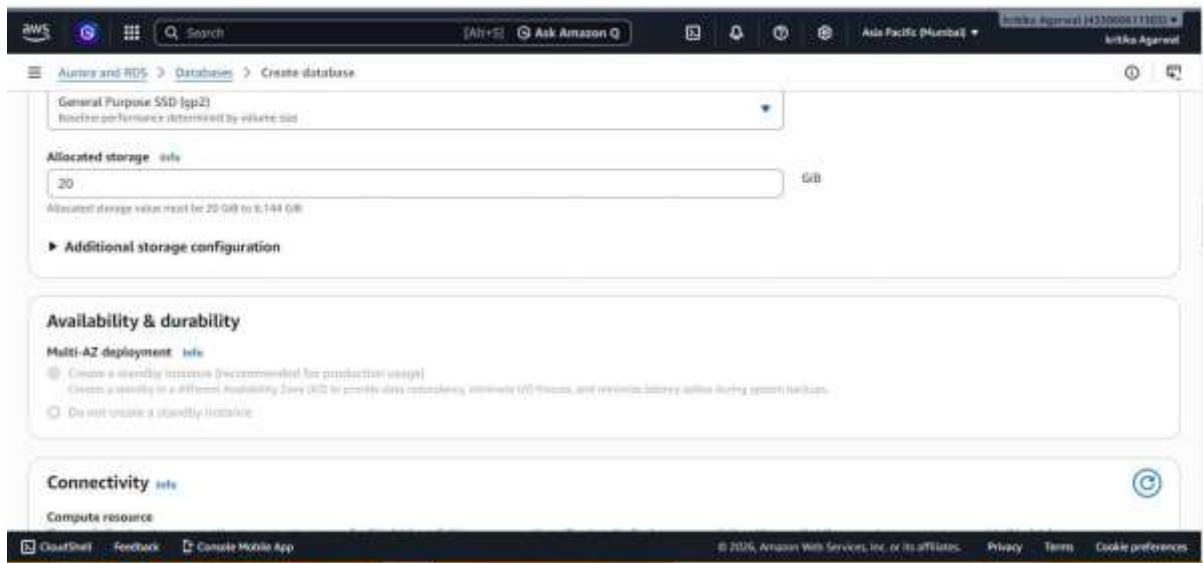


FIG-3:STORAGE

STEP6:Security Group-

- Enabled Public Access.
- Allowed Port 3306 in the Security Group.

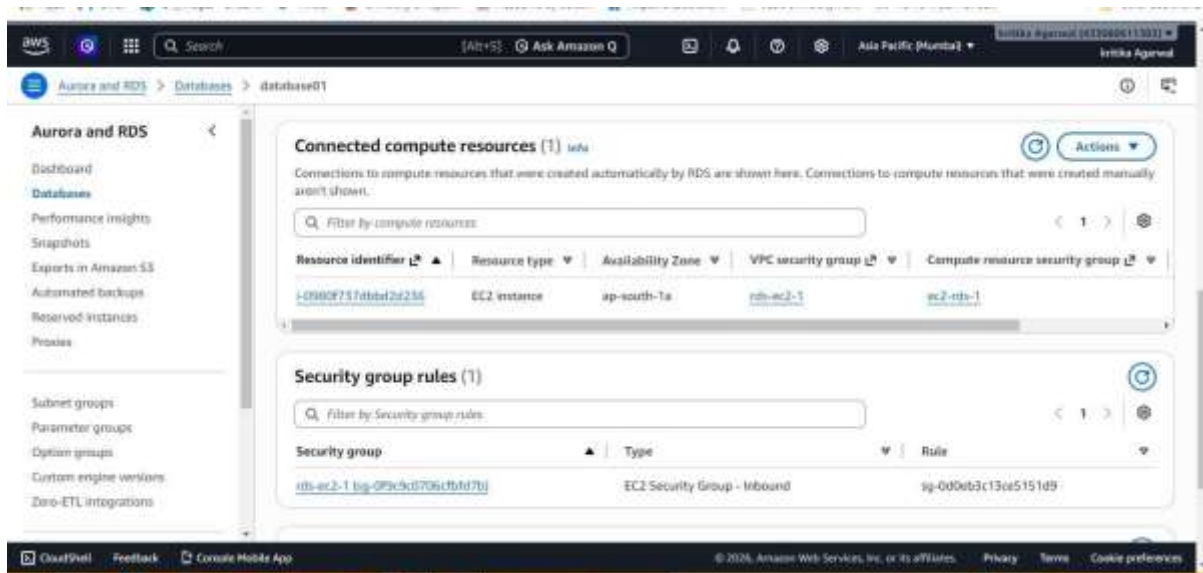


FIG:SECURITY GROUP

STEP7:Connectivity-

- Select VPC.
- Select/Create Security Group.
- Allow Port 3306 (MySQL/MariaDB).

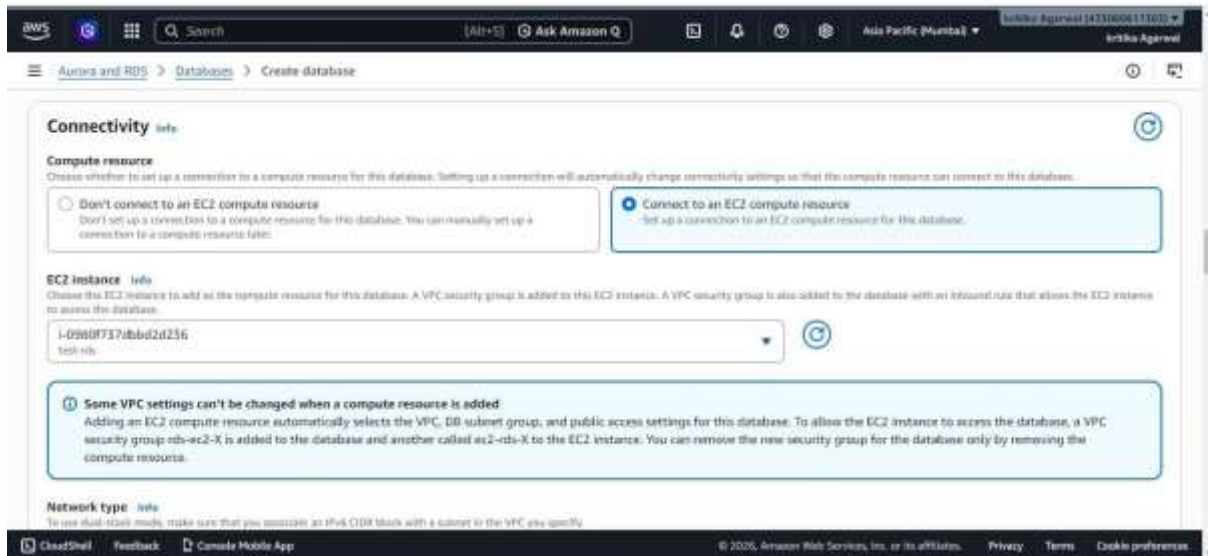


FIG-1:CONNECTIVITY

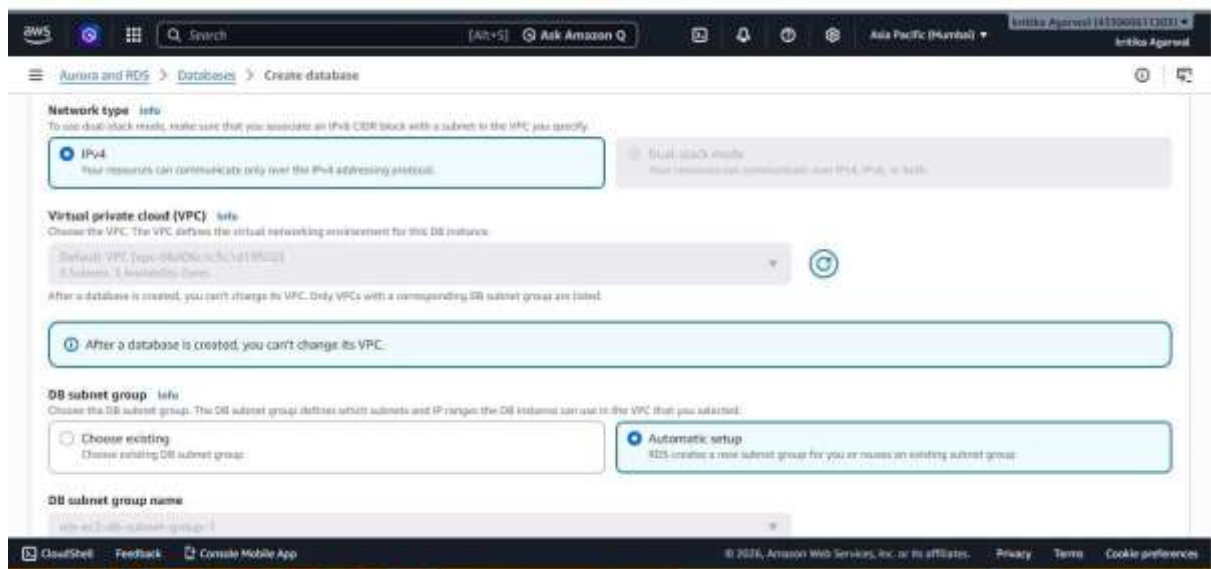


FIG-2:VIRTUAL PRIVATE CLOUD

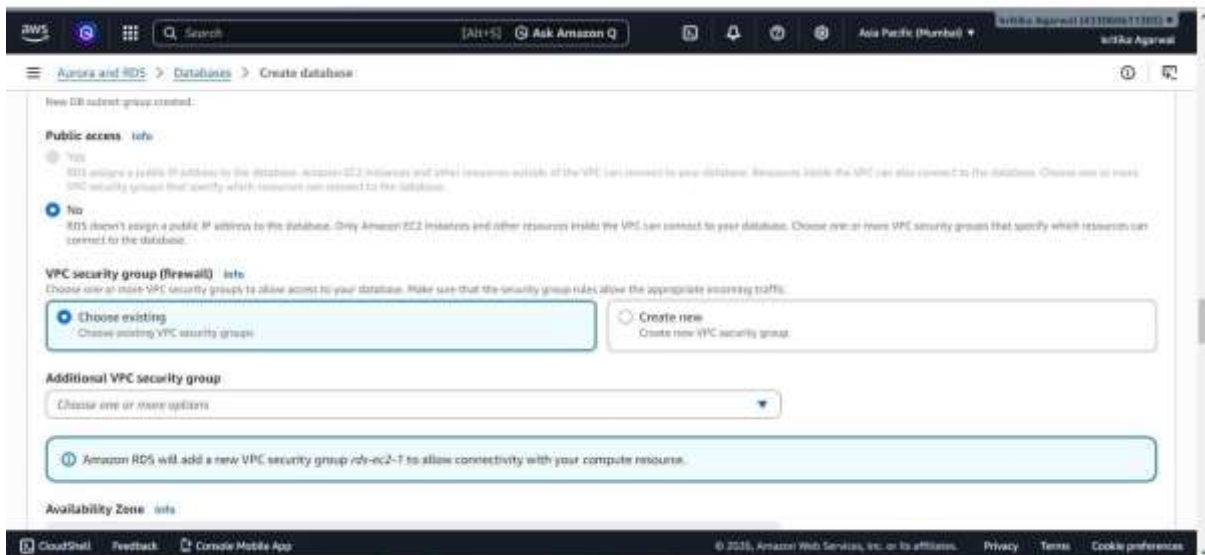


FIG-3:PUBLIC ACCESS

STEP8:Additional Configuration-

- Keep remaining settings as default.
- Click Create Database.

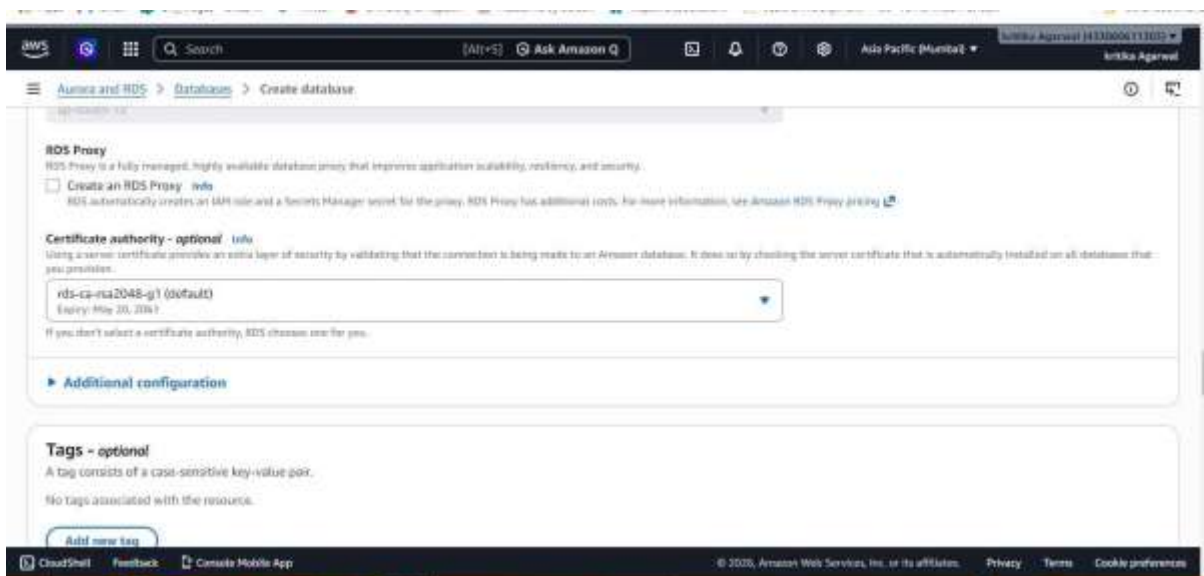


FIG-1:ADDITIONAL DATA

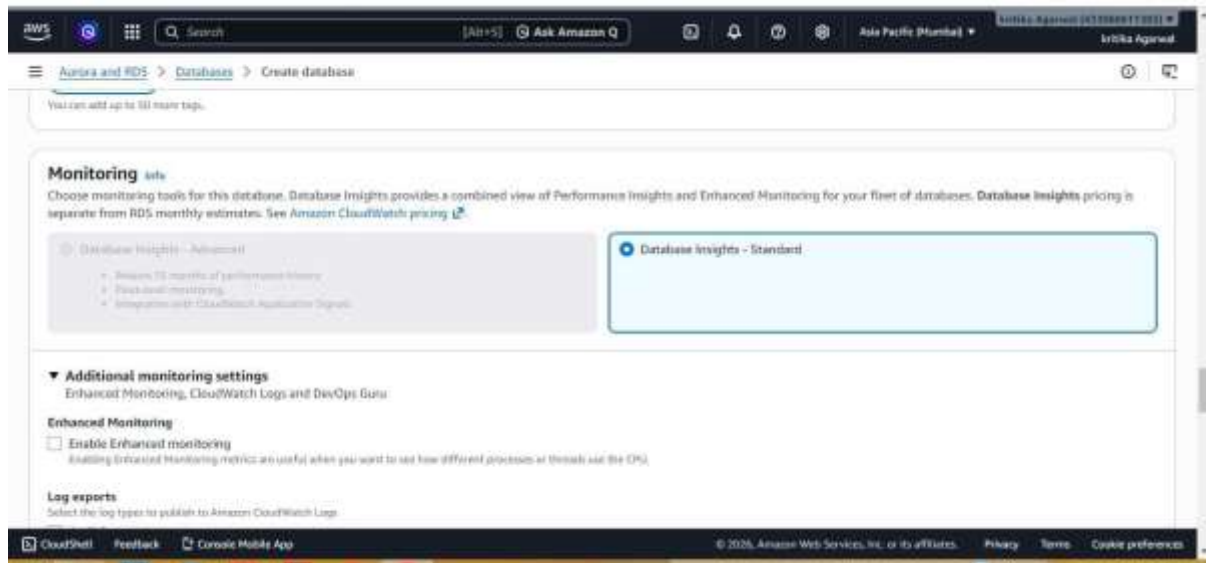


FIG-2:MONITORING

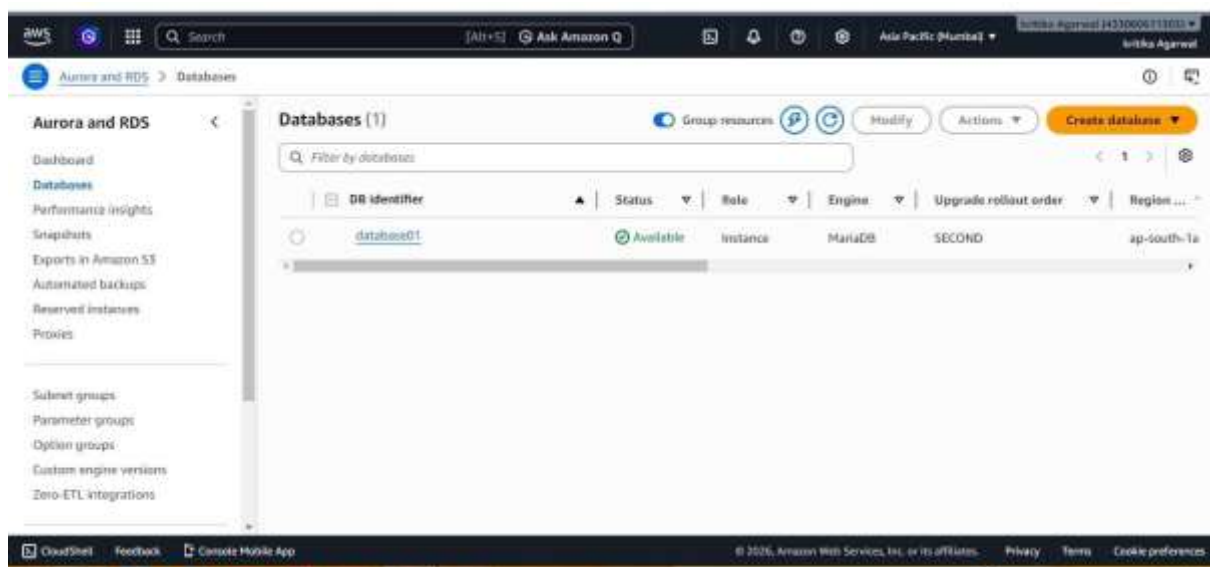


FIG-3:CREATE DATABASE

STEP9:Wait For Availability-

- Wait until the database status becomes Available.
- Copy the RDS Endpoint.

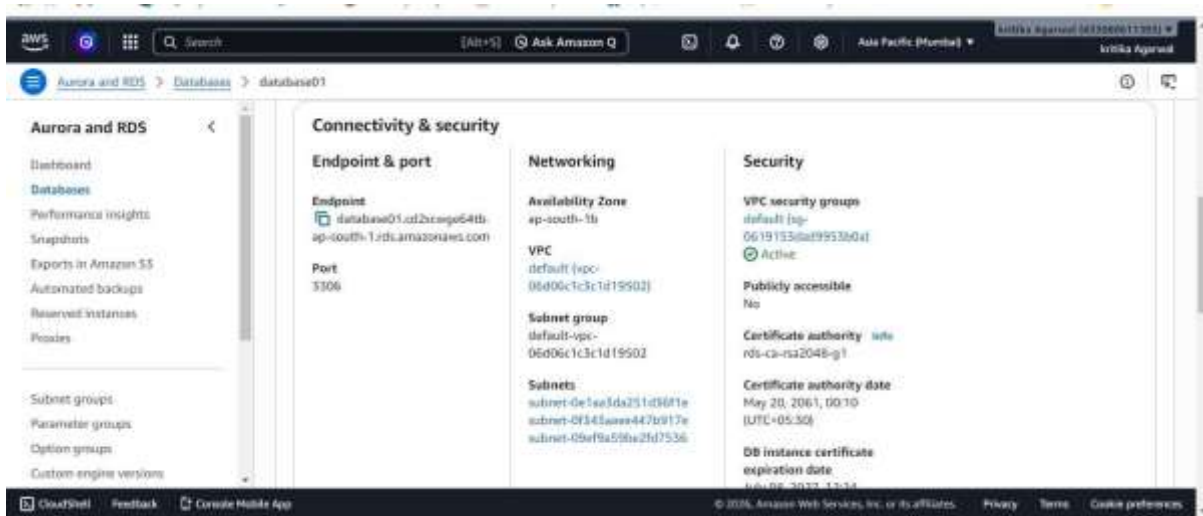


FIG: COPY ENDPOINT

4. IMPLEMENTATION STEPS:

STEP1: Connect from EC2 Terminal-

- Install MariaDB/MySQL client:-
- `dnf install -y mariadb-server`
- `Systemctl status mariadb`
- `Systemctl enable --now mariadb`

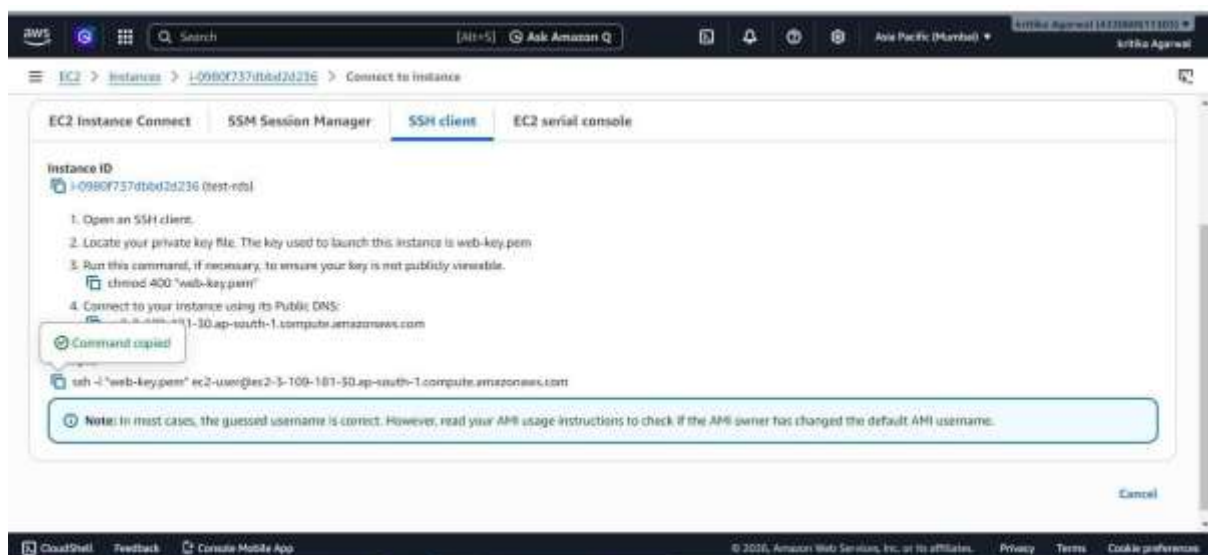


FIG-1:CONNECT SSH

```
root@RDS:~#
The RDS client and Red Hat Lightspeed will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://rhc.rh/registration
Last login: Tue Jul 7 09:04:14 2020 from 157.46.99.246

[ec2-user@RDS ~]$
[ec2-user@RDS ~]$ sudo -i
[ec2-user@RDS ~]$ hostnamectl set-hostname RDS
[ec2-user@RDS ~]$ sudo
[ec2-user@RDS ~]$ sudo systemctl --no-reload enable mariadb-server
[ec2-user@RDS ~]$ systemctl status mariadb
● mariadb.service - MariaDB 10.11 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset; disabled)
   Active: active (running) since Tue 2020-07-07 09:04:04 UTC; 20min ago
   Invocation: 4f460460327447300604f7a5b34831
   Main PID: 2220 (mariadbd)
   CGroup: /system.slice/mariadb.service
           └─ 2220 /usr/bin/mariadbd --basedir=/var

Jul 07 09:05:04 RDS mariadb-prepare-db-dir[2270]: you need to be the system 'mysql' user to connect.
Jul 07 09:05:04 RDS mariadb-prepare-db-dir[2270]: After connecting you can set the password, if you would need to be
Jul 07 09:05:04 RDS mariadb-prepare-db-dir[2270]: able to connect as any of these users with a password and without sudo
Jul 07 09:05:04 RDS mariadb-prepare-db-dir[2270]: See the MariaDB Knowledgebase at https://mariadb.com/kb
Jul 07 09:05:04 RDS mariadb-prepare-db-dir[2270]: Please report any problems at https://mariadb.org/t/r/a
Jul 07 09:05:04 RDS mariadb-prepare-db-dir[2270]: The latest information about MariaDB is available at https://mariadb.org/.
Jul 07 09:05:04 RDS mariadb-prepare-db-dir[2270]: Consider joining MariaDB's strong and vibrant community:
Jul 07 09:05:04 RDS mariadb-prepare-db-dir[2270]: https://mariadb.org/get-involved
Jul 07 09:05:04 RDS [mariadbd][2220]: mariadb.service: Referenced but unmet environment variable evaluation to an empty string: MYSQLOPTS, _MREP_NEW_CLUSTER
Jul 07 09:05:04 RDS systemd[1]: Started mariadb.service - MariaDB 10.11 database server.
[ec2-user@RDS ~]$
```

FIG-2: ENABLE MARIADB SERVICE

STEP2:Connect to RDS-

- mysql -h <RDS-Endpoint> -u admin -p
- Enter the master password.

```
root@RDS:~#
[ec2-user@RDS ~]$ mysql -h database-01.c2scvge64tch.ap-south-1.rds.amazonaws.com -u admin -p
Enter password:
mysql: [Warning] Access denied for user 'admin'@'172.31.37.211' (using password: YES)
[ec2-user@RDS ~]$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 5
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> ALTER USER 'root'@'localhost' IDENTIFIED BY 'Root@12345';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]> EXIT;
bye
[ec2-user@RDS ~]$ mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 6
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

FIG:RDS ENDPOINT (SUCCESSFUL LOGIN)

STEP3:Create Database & Verify-

- CREATE DATABASE database01;
- SHOW DATABASES;

```
root@RDS:~#
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.
Type 'help;' or '?' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show databases
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
# rows in set (0.001 sec)

MariaDB [(none)]> exit
bye
[root@RDS ~]# mysql -u root
ERROR 1045 (28000): Access denied for user 'root'@'localhost' (using password: NO)
[root@RDS ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MariaDB connection id is 55
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.
Type 'help;' or '?' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show databases
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
# rows in set (0.000 sec)

MariaDB [(none)]>
```

FIG:DATABASE VERIFY

OUTPUT:

- MariaDB RDS instance created successfully.
- EC2 connected to the RDS database.
- Database database01 created successfully.

5.CONCLUSION:

SUMMARY-

The AWS RDS MariaDB instance was successfully created and connected to the EC2 instance. The database was accessed successfully, and basic SQL operations were performed.

LEARNING OUTCOME-

- Learned how to create an AWS RDS MariaDB instance.
- Learned to configure VPC and Security Group for database access.
- Learned to connect EC2 with RDS using the endpoint.
- Learned basic SQL commands for database creation and verification.

THANK YOU
